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# Fast and Efficient Image Retrieval via Fully-Convolutional Hashing Network

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***Abstract***

**With the rapid development of information technology, content-based image retrieval and related technologies have become increasingly important. The hash method can represent an image with a sequence of binary codes consisting of 0 and 1, while the application of a convolutional neural networks can learn directly from the image to a discriminative binary code. We propose a deep hash algorithm based on full convolutional neural network, which can reduce the number of network parameters and have the retrieval performance not lower than the cutting-edge method in this field. The proposed network structure uses convolutional layers and the global average pooling layer to replace fully-connected layers in current deep hashing network structures, which significantly reduces the complexity of the network and improved its training performance. This method is called Fully-convolutional hashing networks (FCHN), and experiments were carried out on several publicized datasets to verify the effectiveness of the method.**

***Keywords—deep hashing, fully-convolutional neural networks, image retrieval***

## I. INTRODUCTION

With the rapid development of information technology, Internet is becoming the main platform of transmitting various forms of information, especially images and videos. At the same time, it causes the need of applications about image recognizing and retrieval. Not only search engines for content based image retrieval, but also the use of black and white name lists to identify images transferred, are both in need of using features to describe images properly, in order to retrieve fast and accurately. So content based image retrieval gets widely concentration from academia and industry.

Content based image retrieval normally includes the extraction, processing and matching of image features, because of the abundant information included by digital images, features extracted usually costs lots of time to take out nearest neighbor matching, which makes it cannot cover the need of large-scale image retrieval. On the other hand, image hashing can describe an image with a binary code only consists of 0 and 1, by replacing Euclidian distance with bit operation when doing nearest neighbor matching[13,14], it has an obvious speed advantage when dealing with this kind of tasks. Data-independent image hashing achieved remarkable achievements when convolutional neural networks(CNN) is not widely applied in the field of image recognition, such as LSH[1], SpH.[11]

As the increment of the scale of data that image retrieval tasks faces, traditional image hashing methods based on mapping cannot satisfy the need of large-scale datasets. At the same time, CNN was proposed and achieved remarkable improvements on image related tasks such as classification and detection[27,28,29]. After proposed, CNN shows remarkable ability to learn features represents images, and its performance is still increasing by the proposition of new CNN architectures[34,35].Hashing methods based on deep learning was proposed soon after CNN. These data-dependent methods has better encoding schemes comparing to data-independents methods[19,20,21], which projects indexes of images to manually constructed ANN search by compact hash codes.

Based on CNN deep hashing can learn binary codes describing images from them directly. Deep hashing networks can be trained end-to-end to use proper non-linear functions for hashing coding. These methods achieved state-of-art performance until now[6,7,8,22,23,24,25].

The performance of a deep hashing method relies on its network structure, training process and loss function. Previous researchers mainly focus on these aspects[6,7,8,31,32,33]. We also address this problem with a novel network structure to improve the effect based on current deep hashing methods.

Most current deep hashing methods use fully-connected layer and a latent layer adjacent to it to produce hash codes.

The fully-connected layers are the same to these in AlexNet[2] to transform features extracted by convolutional and pooling layers before to a vector representing the feature of the input image. Then the latent layer implements partial connections between it and the fully-connected layer before, to produce the hash codes[26]. This kind of methods can learn image representations effectively since training weights of fully-connected layers can guarantee all information of the input image is utilized. However, through the analysis on the number parameters of every layer, we can denote that they are mainly from fully-connected layers.

One of the main breakthrough from Multi-layer Perceptron(MLP) to CNN is parameter sharing between convolutional cores on a feature map. This means during the training stage, the network has no need to get every net form nodes of one layer to the other after it, which greatly decreases the number of parameters, as well as making the whole network much easier to train. Another improvement of convolution layers comparing to MLP is local perception. This design is inspired by the way human eyes recognize images and objects, which is from partial perception to an overall acknowledgement. To the image itself, the connection between a pixel to another one near to it is closer than the one far from it. Fully convolutional networks already got improvements dealing with other aspects of image related tasks, such as image segmentation[3] and image classification[4].

Inspired by this , we proposed our novel deep hashing model generates hashing codes that can represent the information of the input image, and can do image retrieval

tasks efficiently and accurately in Hamming space. The model is named Fully-convolutional hashing networks(FCHN). We use convolution layers and global average pooling layer to replace fully-connected layers and latent layer which is local connected to the fully-connected layer before to produce hash codes. The proposed FCHN model can be trained end to end through back-propagation. By using global average pooling layer, this model can preserve global information of the input image better. We took out experiments to prove FCHN can generate compact hash codes that can do image retrieval tasks well and get state-of-art results on two datasets: CIFAR-10 and Caltech-101[12].

## II. RELATED WORKS

Recently, with the application of deep learning methods in the field of computer vision, deep hashing get breakthrough results on various image retrieval datasets[15,16,17,18]. CNNH[6] is the world's first deep hashing method proposed supervised deep learning to generate binary codes from an image. It used a similar matrix labelling the similarity between image pairs and minimize the softmaxloss to train the network. Although it got great improvement comparing to hashing based on hand-craft features, it cannot train the network end to end, which makes it was just using a CNN to extract features.

DNNH[7] is the first deep hashing method to train the network end-to-end. It proposed the latent layer using the output of fully-connected layer as the source of binary codes, from these two layers, it applied local connection to make a certain part of fully-connected layer output nodes correspond to a bit of the final binary code. Another improvement of DNNH is triplet input, the input contains three images, the first image is similar to the second, while it is dissimilar to the third. By minimize the Hamming distance between the first and second one, and maximize the distance between the first and the third, the network could be trained.

HashNet[8] improves the training process of deep hashing. Since the constraint function of hashing is discrete, former deep hashing methods use activation functions like sigmoid to constraint the output of latent layer, then get binary codes by doing a threshold operation. Different from this, HashNet use a sequence of tanh activation function to approach the discrete constraint condition by training the whole network and iteration.

Previous methods applied improvements of all aspects we mentioned before, but they keep using fully connected layer to generate binary codes. This kind of structures takes a large number of parameters by using fully connected layer, which makes a great waste of computation resources, as well as making the network tough to train. We propose a novel fully-convolutional deep hashing network structure to reduce the number of parameters to almost a magnitude less and keep the retrieval performance being competitive comparing to state-of-art methods. To our knowledge, we are the first to apply this structure to image retrieval tasks of deep hashing field.

## FULLY CONVOLUTIONAL HASHING NETWORKS

FCHN is a supervised deep learning method, that means we use a labelled training set to train the deep hashing network. Similar to previous retrieval method, we are given a training set of N samples $\{x_i\}_{i=1}^{N}$, every item $x_i$ refers to a D-dimentional feature vector. At the same time, from the class label data of image dataset, we can infer the label of every input image. Our model get a single image as input and generate binary code for Hamming space retrieval. We can get the hash code through an end-to-end pipeline:(1) several convolution and max-pooling layers to extract image features.(2) a global averaging pooling layer and a sigmoid loss function to produce binary codes.(3)a softmax-loss function to train the whole network

### A. Deep Architecture design

At common time, CNN will append several fully-connected layers after convolution layers to convert a feature map to a vector prepare for the classification or matching the net stage. Because there is a weight parameter between every output node of the former layer and the input node of the latter layer, the parameter scale could be huge.

Table 1 The number of parameters in AlexNet

| Layer name | Layer information | Parameter number |
|---|---|---|
| Input | 224×224×3 | |
| Conv1 | 96×11×11×3 | 34848 |
| Conv2 | 256×5×5×48 | 307456 |
| Conv3 | 384×3×3×256 | 885120 |
| Conv4 | 384×3×3×192 | 663936 |
| Conv5 | 256×3×3×192 | 442624 |
| FC1 | | 37752832 |
| FC2 | | 16781312 |
| Output | | 4097000 |
| total | | 60965182 |

From Table 1 we can infer that a typical AlexNet[2] structure has about 60 million parameters, among them more than 50 million are from fully-connected layers. Although comparing to MLP networks, AlexNet has just two fully connected layers remained, these two layers contains most of the parameters. This caused the difficulty of training and the risk of overfitting. From replacing them by convolutional and pooling layers, we can reduce the number of parameters significantly, as well as improve related performance of the network.

From the application of fully-convolutional networks on image segmentation, we can infer that there's some association between convolution layers and fully-connected layers in a CNN. Actually, the only difference between them is convolution layers only connect a local area the feature map before and share weight between different areas. But it preserves the ability to extract image features. So we use it to replace fully-connected layers to compress feature map. To the latent generate hash code through local connection, we can use global average pooling layer.

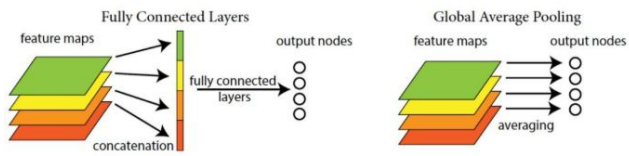


Figure 1 Comparison between fc layer and global average pooling layer

Global average pooling layer was proposed in 2014[9], from figure 1 we can get the difference between fully connected layers and global average pooling layer. The former one connect the input feature map to calculate the output vector of correspond node. Although it ensures feature map input can get abundant learning, it severely increase the calculation complexity.

On the other hand, although the output of global average pooling layer is also an one-dimension layer, each element in this vector is calculated from the average pooling operation of each layer of the input feature map. To a input feature map of dimensions as w×w×c, it will convert every w×w feature to 1×1, so the output is 1×c. By decrease the number of output nodes, it significantly reduces the number of parameters.

Because of these advantages, fully-convolutional neural networks have replaced classic CNNs in some computer vision tasks such as image segmentation and object detection. But in current deep hashing methods, researchers still keep using fully-connected layers. This is because hash codes can be produced by the partially-connected mechanism between the fully-connected layer and the latent layer. However, by adding convolutional layers and pooling layers we can decrease the scale of feature map step by step, to make the input of global average pooling layer keep enough information of the input image.

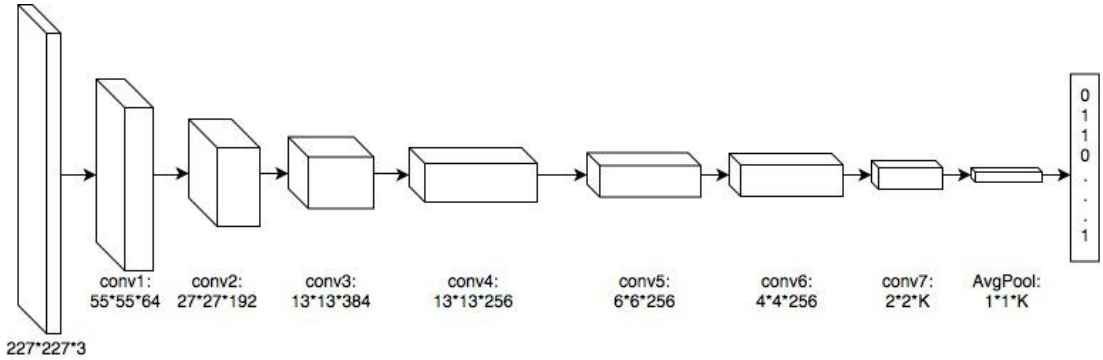


Figure 2 The structure of FCHN

Table2 parameter number of FCHN

| Layer name | Input channels | Conv size | Output size | Number of parameters |
|---|---|---|---|---|
| Conv1 | 3 | 11×11 | 55×55×64 | 23296 |
| Conv2 | 64 | 5×5 | 27×27×192 | 230592 |
| Conv3 | 192 | 3×3 | 13×13×384 | 885120 |
| Conv4 | 384 | 3×3 | 13×13×256 | 663936 |
| Conv5 | 256 | 3×3 | 6×6×256 | 442624 |
| Conv6 | 256 | 3×3 | 4×4×256 | 442624 |
| Conv7 | 256 | 2×2 | 2×2×128 | 131200 |
| OutPut | | | 1×1×128 | 128000 |
| Total | | | | 2947392 |

From all above, we proposed our deep architecture as figure 2. Our deep hashing networks contains 7 convolutional layers to process the input image and get its feature, after the first, second, fifth convolutional layer, we have a max-pooling layer to decrease the scale of feature map. then we use a convolution layer to process the depth of feature map to K, K refers the number of bits in the output binary code. After it, we use a global average pooling layer to process it to a vector of length K. Then after a sigmoid function as the constraint, and a threshold operation, we can get the binary code.

From table 2, we can get the structure of our network in details. Comparing to other deep hashing networks and the original AlexNet structure, our FCHN significantly decreases the number of parameters. The number of total parameters is decreased by nearly a magnitude.

*Training and testing*

Our method uses softmax function and cross entropy to calculate the loss and train the whole network by back propagation. Softmax function can project multi outputs of nodes to (0,1) and make the sum of possibility of an input image to be predicted as all classes at 1, to interpret a multiclass classification. The possibility of one input $x_i$ as label j is:

$$p_j(i) = \frac{e^{z_{ij}}}{\sum_{l=1}^{L} e^{z_{il}}} \quad (1)$$

$z_i$ is the classification result output by last layer, L is the class number in total. During forward propagation, this layer picks the j of max $p_j(i)$ as the class predicted for current input. Corresponding cross-entropy loss function is:

$$J(\theta) = -\frac{1}{n}\sum_{i=0}^{n}\sum_{j=0}^{d} I(label(i) == k) lnP_i \quad (2)$$

$I(label(i) == k)$ is the indicate function to indicate whether the predict class is matching with the ground truth value. If the ground truth of the i[th] input is k, the value of I is 1, otherwise is 0. As the same with Variance loss function, the cross-entropy loss function has characteristics of non-negative and when the predicted result approaches the ground truth, the value of cross-entropy loss function will approach 0.So the optimization function of the whole model is to minimize $J(\theta)$.Furthermore, accomplish with sigmoid activation function, the cross-entropy loss function can overcome small gradient problem in most of input range.

To the retrieval test, we use the class label of images to construct similarity matrix S where $s_{ij} = 1$ refers $x_i$ and $x_j$ are from the same class, we use the Hamming distance to get the similarity between two hash codes $\boldsymbol{h_i}$,$\boldsymbol{h_j}$ ,for every query image $x_i$,get $x_j$ to minimum $d(\boldsymbol{h_i},\boldsymbol{h_j})$ , the function $d(\boldsymbol{h_i},\boldsymbol{h_j})$ calculates the hamming distance between the two hash codes by a bitwise and operation. Then check whether $s_{ij} = 1$ to get the retrieval result.

## III. EXPERIMENTS

In order to check the performance of FCHN in image retrieval tasks, we took out experiments comparing with several state-of-art shallow or deep hashing method on two datasets.

We compare the retrieval performance with classical hashing method LSH[10],SpH[11],ITQ-CCA[12],and Deep hashing methods CNNH[7],DNNH[8],HashNet.[9]To classical hashing methods, we use SIFT(Scale-invariance feature Transform) [30] features to get the representation of an input image, then generate binary codes through these methods. To deep hashing methods, we choose the network generating binary codes based on AlexNet to implement them.

We did Hamming space retrieval on these datasets:(1) get the binary code of query image (2) return the image in dataset with the minimum Hamming distance. (3)check the ground truth whether they are similar images.

### A. Datasets

**CIFAR10**: includes 60000 images from 10 categories. During the experiment, we divide the dataset at 1:1:4 to test, validation and train set.

**Caltech101[12]:** includes approximately 8500 images from 101 item classes and 1 background class.

### B. Results

| | Cifar10 | Caltech101 |
|---|---|---|
| LSH | 0.2858 | 0.2585 |
| SpH | 0.3016 | 0.2711 |
| ITQ-CCA | 0.3668 | 0.3212 |
| CNNH | 0.5520 | 0.5174 |
| DNNH | 0.6474 | 0.6048 |
| HashNet | 0.7702 | 0.7051 |
| FCHN | **0.8276** | **0.7176** |

Table 3 The experiment result

From table3 we can get the retrieval results on these two image datasets. From this table, we can infer that the results of deep hashing methods are significantly better than traditional ones. Among the deep hashing methods, Our FCHN can reach the state-of-art performance on these datasets. And in cifar10, our advantage is more obvious, because of the convolution layer to reduce the scale of feature map extracted from the input image and the global average pooling layer to make each binary codes replace a layer from the original feature map, FCHN is using global feature rather than local feature.

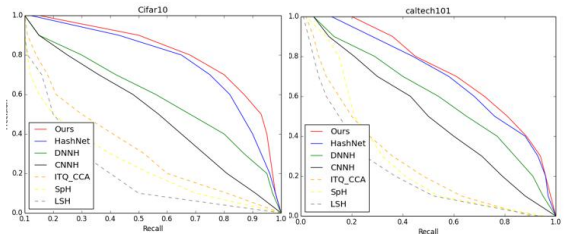


Figure 3 The precision-recall curve

Figure 3 is the precision-recall curves figure for these methods on two datasets.From these figures, it is obvious that deep hashing methods has better performance than traditional ones, and the result of out method is better than the others.

## CONCLUSION

This paper proposed a novel and effective deep hashing mothod based on fully-convolutional neural networks.We reviewed the architeture of current deep hashing method, and analysed the number of parameters of each layer from AlexNet. By replacing fully-connected layers in current deep hashing networks, we significantly decrease the number of parameters in a net, as well as decreasing the difficulty of caculation and training.Futher more, by adding convolution and global averaging pooling layers ,we can preserve global information of images better. At last, we carried out experiments on two datasets comparing with state-of-art deep or traditional hashing methods, proved the effectiveness of our model.Specifically the results on cifar10 proves the method we proposed can preserve the global infromation of input images and accomplish the image retireval tasks fast and accurately.

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